Squares with Square Second Arithmetic Derivative: Infinite Families and an Exact Extension of OEIS A348425

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Abstract

OEIS A348425 consists of the nonnegative squares whose second arithmetic derivative is a square. We consolidate 9,114 consecutive terms, compared with the 33-term public snapshot acquired on October 22, 2026. The final campaign contributes 8,917 terms and completes the input-root interval through 10^{10}. It follows a 197-term prefix whose entire root interval has been separately checked. Its arithmetic kernel uses $D^2(r^2) = rD(r) + 2D(r)^2 + 2rD(D(r))$, reducing the irregular factorization operand to $D(r)$. Complete coverage rests on this exhaustive producer and its sealed records; separate arithmetic checks verify all reported positive witnesses and bounded windows. No independent full re-enumeration of the final campaign interval is claimed. We prove a necessary and sufficient square criterion for each fixed reduced core and integer-rate multiplier class, using the historical classification of $D(t) = \lambda t$. Four families established in earlier project notes are recovered, and 31 further certified classes are extracted from the existing data. These constructions give unconditional infinitude without assumptions on primes represented by quadratic polynomials. An adjacent witness already present in an acquired prime table sharpens the next-term bound to $10^{20} < a(9115) \leq 100196144067026155201$; it does not determine that term.

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1 Introduction

The arithmetic derivative is determined by $D(p) = 1$ for every prime p, $D(0) = D(1) = 0$, and the product rule

$$D(uv) = uD(v) + vD(u). \tag{1}$$

Its factorization formula and elementary bounds are classical [1, 2]; a recent exposition is given by Tossavainen et al. [4]. Throughout this paper, $D^2(n) = D(D(n))$ denotes composition, whereas $D(n)^2$ denotes an ordinary square. The function D is not additive.

Marius A. Burtea introduced OEIS A348425 in October 2021 [6]:

$$\mathcal{A} = \{r^2 : r \in \mathbb{Z}_{\geq 0},\ D^2(r^2) \text{ is a square in } \mathbb{Z}_{\geq 0}\}. \tag{2}$$

We write $a(1), a(2), \ldots$ for its increasing enumeration. Thus $a(1) = 0$, $a(2) = 1$, $a(3) = 4$, and $a(4) = 49$ (offset 1); for example, $D(49) = 14$ and $D(14) = 9$. The input root r, the sequence index i, and the second-derivative root $R = \sqrt{D^2(r^2)}$ are different quantities.

The snapshot acquired on October 22, 2026 contains 33 indexed terms. Its last value is $31618129 = 5623^2$. It also contains root-enumeration programs by Amiram Eldar and Michel Marcus and three prime-power constructions [6]. We retain that precise baseline when reporting an extension; the final campaign alone did not produce all 9,081 terms beyond it.

There are two complementary questions. Exact indexing requires checking every intervening input root. A structural family supplies infinitely many members, but cannot establish their sequence indices without that coverage. The first question is addressed by the completed enumeration and its validation. For the second, a product-rule criterion transfers the historical integer-rate multipliers of Kovič [3, Proposition 2] to square second derivatives. The corresponding reduced-core decomposition is already present in Ufnarovski and Åhlander [2, Section 8].

The contributions are an exact local extension to 9,114 terms; proofs and disjointness statements for scale families; a reproducible finite analysis of prime powers and reduced cores; and a tighter, unindexed witness bound. The four initially supplied scale families predate the final campaign: their mechanism was developed in project notes on October 5, 2026, with the fourth seed retained in subsequent notes. The 31 additional classes below are identified by analyzing the consolidated data. We make no claim of absolute bibliographic priority for these specializations. In particular, the book of Balzarotti and Lava [5] has not been consulted in full.

2 Exact identities and scale families

For $n \geq 2$ with prime factorization $n = \prod_p p^{e_p}$, the defining rules give

$$D(n) = \sum_{p \mid n} e_p \frac{n}{p}. \tag{3}$$

We use the empty product convention for 1 and treat 0 separately.

Proposition 2.1 (Root reduction)**.** *For $r \in \mathbb{Z}_{\geq 0}$, put $d = D(r)$ and $e = D(d)$. Then*

$$D(r^2) = 2rd, \qquad D^2(r^2) = rd + 2d^2 + 2re. \tag{4}$$

Proof. The first identity follows from (1). For the second, differentiate the product $2(rd)$, using $D(2) = 1$:

$$D(2rd) = rd + 2D(rd) = rd + 2\big(d^2 + rD(d)\big).$$

This uses the product rule twice; it does not differentiate a sum term by term. Both identities hold also for $r = 0, 1$. □

For prime r, the result is $D^2(r^2) = r + 2$. Prime input roots are therefore essential, and may not be discarded. More generally, the identity replaces factorization of $2rD(r)$ by factorization of $D(r)$.

Lemma 2.2 (Integer-rate multipliers)**.** *Let $\lambda \in \mathbb{Z}_{\geq 0}$. The positive integer solutions of $D(t) = \lambda t$ are exactly*

$$t = \prod_p p^{pc_p}, \qquad c_p \in \mathbb{Z}_{\geq 0}, \qquad \sum_p c_p = \lambda, \tag{5}$$

where the support is finite. In particular, $\lambda = 0$ forces $t = 1$.

Proof. This is the integer specialization of the classical logarithmic decomposition [2] and Kovič's Proposition 2 [3]. The case $t = 1$ has $D(t) = 0$ and $\lambda = 0$, and satisfies the empty-product convention. For $t > 1$, write $t = \prod_{p|t} p^{e_p}$. Then $\sum_{p|t} e_p/p = \lambda$. Multiply by $P = \prod_{p|t} p$ and reduce modulo a fixed prime $p \mid t$. All terms except $e_p P/p$ vanish, so $p \mid e_p$. Setting $e_p = pc_p$ proves necessity; substituting in (3) proves sufficiency. For $\lambda = 0$, all $c_p = 0$, so $t = 1$. □

Equivalently, t is a product of λ factors p^p, with primes allowed to repeat. There is no coprimality restriction between these primes and any separately chosen seed.

Theorem 2.3 (Square criterion in a scale class)**.** *Let $m, t \geq 1$ and $\lambda \in \mathbb{Z}_{\geq 0}$ satisfy $D(t) = \lambda t$. Define*

$$b = D(m) + \lambda m, \qquad C(m, \lambda) = 2D(mb) + (4\lambda + 1)mb. \tag{6}$$

Then

$$D^2((mt)^2) = t^2 C(m, \lambda). \tag{7}$$

Consequently $(mt)^2 \in \mathcal{A}$ if and only if $C(m, \lambda)$ is a square. If $C(m, \lambda) = S^2$, its second-derivative root is St.

Proof. The product rule gives $D(mt) = tb$, hence $D((mt)^2) = 2t^2 mb$. Moreover, $D(t^2) = 2\lambda t^2$ and $D(2t^2) = (4\lambda + 1)t^2$. Differentiating the product $(2t^2)(mb)$ proves (7). The coefficient is a nonnegative integer. Multiplication by t^2 preserves every prime-exponent parity, so it is a square exactly when the product is. The only zero case is $m = 1$, $\lambda = 0$, hence $t = 1$, giving the member 1. □

Definition 2.4. For $r \geq 1$, write each prime exponent uniquely as $e_p = pc_p + b_p$, with $0 \leq b_p < p$, and set

$$m = \prod_p p^{b_p}, \qquad t = \prod_p p^{pc_p}, \qquad \lambda = \sum_p c_p. \tag{8}$$

We call m the reduced core and λ the canonical rate of r. A positive integer m is *reduced* when $\nu_p(m) < p$ for every prime p, where ν_p is the prime exponent.

This is the small/large-parts decomposition of [2, Section 8], expressed in input-root notation. It is unique, $r = mt$, and Lemma 2.2 applies to t. Thus Theorem 2.3 is an exact criterion for each core and rate, but does not solve all possible conditions $C(m, \lambda) = S^2$.

Corollary 2.5. *If m is reduced, $\lambda \geq 1$, and $C(m, \lambda)$ is a square, then*

$$\left\{ \left(m \prod_{j=1}^{\lambda} p_j^{p_j} \right)^2 : p_1, \ldots, p_\lambda \text{ prime} \right\} \subseteq \mathcal{A}$$

is infinite. Families from distinct reduced-core/rate pairs are disjoint.

Proof. Membership follows from the preceding theorem and lemma. Fix all but one prime parameter, and let the remaining prime increase; the roots are unbounded. For disjointness, multiplication adds multiples of p to the exponent of each prime p, even when p divides m. Since $0 \leq \nu_p(m) < p$, adding these multiples produces no carry from the reduced exponent into the quotient. The residues and the total quotient sum in (8) recover m and λ uniquely. A root cannot belong to two distinct pairs. □

2.1 Four explicit scale families

Table 1 gives exact certificates for the four seeds from the earlier notes. Every seed is reduced: they are 47, 5^3, $5^3 7$, and 23^2. They therefore yield four disjoint infinite families:

$$(47p^p)^2, \quad R = 148p^p, \qquad (125p^p)^2, \quad R = 500p^p,$$
$$(875p^p)^2, \quad R = 3125p^p, \qquad (529p^p q^q)^2, \quad R = 2852p^p q^q.$$

Here p, q are arbitrary primes, including $p = q$, and may divide the seed. No conjecture about polynomial primes is involved.

Table 1: Exact certificates for the inherited families. The notation is that of (6); $C = S^2$.

m	λ	b	$D(mb)$	C	S
47	1	48	5312	21904	148
125	1	200	62500	250000	500
875	1	1525	1546875	9765625	3125
529	2	1104	1438880	8133904	2852

Formulations using the square seed m^2 describe the same families: $D(t^2) = 2\lambda t^2$, so the corresponding multiplier rate for square inputs is 2λ, rather than λ.

2.2 Prime powers and the parameter convention

Proposition 2.6. *For a prime p and $k \geq 1$,*

$$D^2(p^{2k}) = p^{2k-2}\big[pD(2k) + 2k(2k-1)\big]. \tag{9}$$

Thus $p^{2k} \in \mathcal{A}$ exactly when the bracket is a square.

Proof. Differentiate $D(p^{2k}) = (2k)p^{2k-1}$ as a product. The result is $D(2k)p^{2k-1} + (2k)(2k-1)p^{2k-2}$. The remaining factor p^{2k-2} is a square. This calculation does not assume $p \nmid 2k$. □

For $k = 1, 2, 3$, the respective brackets are

$$p + 2, \qquad 4(p + 3), \qquad 5(p + 6). \tag{10}$$

These recover the three constructions in A348425 [6]: $p = s^2 - 2$, $p = s^2 - 3$, and $p = 5s^2 - 6$, respectively, subject to primality of p. They do not prove that the required prime sets are infinite. In particular the third condition follows from $5(p + 6) = u^2$: $5 \mid u$, so $u = 5s$ and $p = 5s^2 - 6$.

The parameter conventions of the related entries differ [10, 11, 12]: A028871 lists the primes $p = s^2 - 2$; A028873 lists the parameters s for which $s^2 - 3$ is prime; A201787 lists the primes $p = 5s^2 - 6$, not their parameters. The third comment in the checked A348425 snapshot uses an A201787 term as if it were s. For its second term, 599, the literal assertion fails because

$$5 \cdot 599^2 - 6 = 1793999 = 19 \cdot 94421.$$

The corrected mathematical statement is: if the prime $p \in$ A201787 has $p = 5s^2 - 6$, then $D^2(p^6) = (5p^2 s)^2$. The live text checked on October 22, 2026 retains that parameter error.

3 Enumeration, arithmetic bounds, and completion

Since $r \mapsto r^2$ is strictly increasing on $\mathbb{Z}_{\geq 0}$, exhaustive root enumeration preserves the order in (2). The producer first computes $D(r)$ in blocks, then applies Proposition 2.1 and an exact integer-square test to every root. No family restriction, primality restriction on r, or empirical exclusion is used.

3.1 Sieve and exact arithmetic

For a block $[L, H]$, initialize a residual equal to each root and a zero derivative accumulator. For each prime $p \leq \sqrt{H}$, strip p^e from the residual and add $e(r/p)$ using the original r. Any final residual greater than 1 is prime, since a composite residual would have a prime factor at most $\sqrt{H}$ already processed. The residual prime q has exponent one: $q > \sqrt{H}$ implies $q^2 > H$. Its contribution is r divided by that residual. This proves that the sieve gives $d = D(r)$ exactly. The cases $r = 0, 1$ are handled explicitly.

The irregular operand d is factored using small-prime division, deterministic word primality based on the Miller–Rabin test [19, 20], and a fixed-seed rho accelerator of Pollard type [18]. This uses Floyd cycle detection, constants $c = 1, \ldots, 32$, and at most 100,000 steps per constant. The primality bases are

$$2,\ 325,\ 9375,\ 28178,\ 450775,\ 9780504,\ 1795265022,$$

whose coverage for integers below 2^{64} is the Sinclair record listed in [22]. The Feitsma–Galway tables [21] supply exhaustive base-2 pseudoprime data for this word range; these large tables were not independently regenerated here. Bases are reduced modulo the candidate and a zero residue is skipped. A finite check of all composite divisors of these bases confirms rejection by the remaining tests, including $14089 = 73 \cdot 193$, which survives the initial small-prime screen. Every factor leaf is checked for primality, and the product of the prime powers must reproduce the operand exactly. Exhausting the bounded rho attempts invokes complete odd trial division; it never turns a factorization failure into a negative result. This fallback ensures exactness, without a uniform practical running-time claim for adversarial operands.

The sieve supplies d; only d needs irregular factorization for each root. The producer does not factor $e = D(d)$. After obtaining e, it computes the two expressions in (4). The residue condition $D^2(r^2) \bmod 16 \in \{0, 1, 4, 9\}$ is necessary only. A floating square-root estimate is corrected with integer products until $q^2 \leq y < (q+1)^2$, where $y = D^2(r^2)$; acceptance requires $q^2 = y$. The decision is therefore integral, rather than a floating-point test.

Lemma 3.1 (Width bounds)**.** *For $U \geq 2$, let $F(U) = \lfloor U \lfloor \log_2 U \rfloor / 2 \rfloor$, with $F(0) = F(1) = 0$. If $0 \leq v \leq U$, then $D(v) \leq F(U)$. For $r \leq B$, set $A = F(B)$ and $E = F(A)$. Then*

$$r^2 \leq B^2, \quad d \leq A, \quad e \leq E, \quad D(r^2) \leq 2BA, \quad D^2(r^2) \leq BA + 2A^2 + 2BE. \tag{11}$$

Proof. For $v \geq 2$, (3) gives $D(v) \leq v\Omega(v)/2$, where $\Omega(v) = \sum_p e_p$. Since $2^{\Omega(v)} \leq v$, $\Omega(v) \leq \lfloor \log_2 v \rfloor$. The integer bound follows, because $v \lfloor \log_2 v \rfloor$ is nondecreasing for integer $v \geq 2$; the cases $v = 0, 1$ follow separately. Applying it first to r and then to d, and using (4), proves (11). This is the integer form of the classical upper bound [1]. □

The supported source capacity is $B = 10^{12}$. Here

$$A = 19500000000000, \qquad E = 429000000000000,$$

$$B^2 = 10^{24}, \quad 2BA = 3.9 \cdot 10^{25}, \quad BA + 2A^2 + 2BE = 1.638 \cdot 10^{27} < 2^{91}.$$

Thus roots and factorization operands fit unsigned 64-bit words, while coordinates and their products use unsigned 128-bit arithmetic, with promotion before multiplication. The second-derivative root is at most 40472212689696. These are conservative capacity bounds, not measurements or coverage through 10^{12}.

3.2 Ordered completion and the executed identity

The completed campaign used Linux Mint 22.3 on x86-64 LP64, GCC 13.3.0, OpenMP with 16 workers, and blocks of 65,536 roots. The processor was an AMD Ryzen 9 7940HS (8 cores, 16 hardware threads), with 64 GB DDR5 installed RAM. The runtime libraries were GNU libgomp (package 14.2.0-4ubuntu2~24.04.1) and OpenSSL 3.0.13; the Python checks used Python 3.12.3 and SymPy 1.12. Appendix A gives the exact build options and record grammar. Within a block each worker writes its own result flag. Positive rows are replayed in increasing root order after the join. A block advances the committed frontier only when all its roots finish; an interrupted or failed block cannot certify negatives. The two sealed segments in Table 2 are contiguous.

Table 2: Completed final-campaign segments, with inclusive input roots. The hit counts sum to 8,917.

Segment	First root	Last root	Hits
0	1000001	5454971456	6792
1	5454971457	10000000000	2125

The author executed the campaign with the historical source `a348425_v2.c`. The public-facing copy is `a348425.c`. Its sole C change is the neighboring-source filename used for identity checking; the Makefile changes that filename and verification paths. The arithmetic, capacity, parallelism, record grammar, and build options are unchanged. The original executed source, Makefile, binary, 197-term baseline, and saved run records are preserved separately, with an exact transformation record. The normalized build is not credited with producing the historical results. These provenance materials are distributed with the project package [24]; the original executed identity remains preserved separately from the normalized public build.

The sum of saved internal elapsed counters is approximately 48 minutes 46 seconds. It excludes part of checkpoint I/O and other startup or closure costs and is not the complete command wall time. A campaign peak-RSS measurement is unavailable; short calibration measurements are not substituted for it. Rho-fallback counts were printed during execution but are not retained in the saved campaign records, so no observed fallback frequency is reported. No unmeasured performance comparison is made.

4 Indexed extension and its validation

Theorem 4.1 (Computational result)**.** *The separately checked prefix and the completed exhaustive producer give 9,114 consecutive terms of $\mathcal{A}$ through $r = 10^{10}$. Coverage of the final interval $1000001 \le r \le 10^{10}$ rests on this producer and its preserved completion records, rather than an*

independent re-enumeration of that full interval. In particular,

$$a(36) = 54582544 = 7388^2,$$
$$a(198) = 1012050084049 = 1006007^2,$$
$$a(9114) = 99948010099131627889 = 9997400167^2.$$

No further member has input root at most 10^{10}*. The first unresolved index is 9115.*

Computational evidence and coverage argument. The chronology is given in Table 3. The original extension admitted the 33-term public baseline. During manuscript revision on October 23, 2026, complete trial division applied directly to r^2 and $D(r^2)$ for every $0 \le r \le 6500$ found exactly the 35 listed members, including exactly those 33 through $r = 5623$. A separately supplied C implementation agreed on this full interval; it factors r and $D(r)$ and evaluates $D(2rD(r))$ from the merged prime multiset. This closes the earlier admitted-prefix dependency, without attributing the later check to the original campaign. Before that revision, an exact replay of 877 roots, $5624 \le r \le 6500$, found the two additional roots 5625 and 5927. Direct trial division and a separate SymPy computation agreed on this whole interval. A further complete finite check of 993,500 roots, $6501 \le r \le 10^6$, gave 162 more members, agreeing with a separate completed producer run on that finite interval and yielding a 197-term prefix. This check used a separately supplied Python implementation: it factors r and $D(r)$ by trial division and forms the prime multiset of $2rD(r)$, rather than evaluating the second formula in (4). The delivered implementation and its local execution record are retained.

The final producer covers exactly $10^{10} - 10^6 = 9999000000$ further roots in the two segments of Table 2. Its committed cursors reach the inclusive upper endpoints, the segments join without gaps or overlap, and the sealed hit lists agree with their ordered aggregate. They contain $6792 + 2125 = 8917$ positives. The preserved prefix and these ordered hits give $197+8917 = 9114$ indexed rows. The root reduction, exact tests, and complete-block commitment establish how the producer accounts for negatives. In particular, the last positive root is not used as a proxy for the completed frontier. □

Table 3: Distinct stages of the local extension, relative to the acquired public baseline. Its completeness was separately checked on October 23, 2026; this later verification adds no terms. The extension beyond the snapshot is $9114 - 33 = 9081$ terms.

Stage	Root frontier	Added terms	Total terms
Public snapshot	5623	—	33
Exact replay	6500	2	35
Complete finite continuation	1000000	162	197
Final campaign	10000000000	8917	9114

4.1 What the separate checks establish

All 8,917 final-campaign positives were recomputed in Python with SymPy 1.12 [23], factoring $n = r^2$ and $D(n)$ directly. Factor products and prime-leaf domains were checked. A documentary table with columns

$$(i,\ r,\ n = a(i),\ D(n),\ D^2(n),\ R)$$

was then checked for every one of its 9,114 rows: 18,228 derivative identities and 18,228 square identities, together with equality to the indexed b-file. This table records witnesses for the contiguous terms; it is not a sparse extension across a missing index.

Four complete windows were also checked with separate arithmetic:

$$[1000001, 1001419], \quad [100000001, 100001000],$$
$$[1000000001, 1000001000], \quad [9999999001, 10000000000].$$

They contain $1419 + 1000 + 1000 + 1000 = 4419$ distinct roots, all negative, in agreement with the run. Positive-path agreement is supplied by the all-positive check. During manuscript revision, the separately supplied C checker also reproduced complete 129-root windows $[1005943, 1006071]$ and $[9997400103, 9997400231]$, containing respectively the first and last campaign hits. A third 129-root window $[5454971392, 5454971520]$ straddles the segment join and is negative. These bounded checks supplement, and do not extend, campaign coverage.

These checks have complementary limitations. Recomputing a reported positive does not detect an omitted positive. Seals, identity checks, aggregate equality and completed cursors establish integrity and the producer's recorded completion, rather than a second independent enumeration of all 9,999,000,000 final-campaign inputs. The 33-term baseline dependency was closed by the complete check described above. The separately implemented finite checker shares the mathematical definition, and implementations supplied by one LLM family and reviews from that family have correlated provenance. Execution claims in external code reviews that lack delivered raw harnesses or logs are not counted as local reruns.

Source review before execution concerns software correctness and does not by itself verify the structural theorems. During manuscript preparation, the current data were again checked by the finite-analysis companion, and the archived original run was verified without advancing it. The provenance checker also regenerated all 9,114 indexed pairs from the original 197-term baseline and the sealed final hit list.

4.2 The first unresolved term

The families already show that $\mathcal{A}$ is infinite, so the next term exists. The completed negative tail gives $a(9115) > 10^{20}$. The earlier family witness

$$r = 529 \cdot 7^7 \cdot 3^3 = 11762664669$$

has square 138360280115340879561 and second-derivative root 63416105172, giving a valid unindexed upper bound.

Inspection of the acquired A028871 table [10] during manuscript preparation on October 22, 2026 identified a tighter witness, also present in the earlier restricted-family list:

$$p = 10009802399 = 100049^2 - 2.$$

Complete trial division through $\lfloor \sqrt{p} \rfloor = 100048$ confirms primality. Proposition 2.1 then gives

$$D(p^2) = 20019604798, \qquad D^2(p^2) = 10009802401 = 100049^2.$$

Consequently

$$10^{20} < a(9115) \leq 100196144067026155201. \tag{12}$$

The provider index of p in A028871 is 8889; it is not an A348425 index. No arbitrary-root interval beyond 10^{10} was enumerated to obtain this improvement. In particular, the witness is not asserted to equal $a(9115)$.

5 Finite structure of the consolidated data

The analysis in this section examines only the existing 9,114 positive rows. Its companion program checks direct composition, factors each input root, constructs (8), and verifies the exact coefficient and root in Theorem 2.3. It searches neither additional roots nor a larger seed domain.

5.1 Prime powers and a disjoint classification

Apart from $r = 0, 1$, there are 8,888 prime roots, 52 higher prime-power roots, and 172 roots with at least two distinct prime factors. The prime-power exponents in $r = p^k$ are given in Table 4. The exceptional higher exponents are 5^8, 17^8 and 3^{20}. They satisfy Proposition 2.6; 5^8 belongs to

Table 4: Prime-power exponents among input roots $2 \le r \le 10^{10}$ of members of $\mathcal{A}$.

k	1	2	3	8	20
Count	8888	45	4	2	1

the inherited core 125, rate 1 family, and 3^{20} to the core 9, rate 6 family below. Thus the three small-exponent constructions do not exhaust the prime-power part.

As a separate provider check, transforming all acquired terms of A028871, A028873 and A201787 within the completed root bound gives respectively 8,888, 45 and 4 members, all present in the consolidated table. These are restricted-domain checks, not coverage for mixed roots. The following disjoint partition classifies membership in the displayed constructions:

$$9114 = 2 + 8937 + 19 + 111 + 45. \tag{13}$$

The summands count, in order, $r = 0, 1$; the $k = 1, 2, 3$ prime-power constructions; the four inherited scale families; the other observed positive-rate classes; and the remaining rate-zero members. Here $8937 = 8888+45+4$. The inherited classes contain respectively 4, 4, 4 and 7 terms. The classification assigns $r = 0, 1$ first, then the $k = 1, 2, 3$ prime-power constructions, then the positive-rate classes, and finally the remaining roots. No summand in (13) counts a term twice. The last 45 terms are not unexplained by the defining predicate: they are outside this particular collection of families.

5.2 Canonical rates and additional certified families

Among the 9,112 roots at least 2, the rate distribution is given in Table 5. There are exactly 35

Table 5: Canonical rates among the 9,112 input roots at least 2.

λ	0	1	2	3	4	5	6	7
Count	8982	28	29	17	3	27	23	3

distinct observed core/rate pairs with $\lambda > 0$, accounting for 130 terms. All their reduced cores are distinct. Table 6 records $S = \sqrt{C(m, \lambda)}$ and the finite number h of terms in the completed domain. Each row gives a universal infinite family by Corollary 2.5, not just its h observed members. The four marked rows are the inherited families; the other 31 are additional certified specializations extracted during this finite analysis, without a bibliographic-first claim.

For example, $m = 9$, $\lambda = 6$ gives $b = 60$ and $D(mb) = D(540) = 1188$, so

$$C(9, 6) = 2 \cdot 1188 + 25 \cdot 540 = 15876 = 126^2.$$

Hence

$$\left(9\prod_{j=1}^{6}p_j^{p_j}\right)^2\in\mathcal{A},\qquad R=126\prod_{j=1}^{6}p_j^{p_j}$$

for arbitrary primes, including repetitions and $p_j=3$. As another example, $m=1847$, $\lambda=1$ has $S=5584$; its $p=2$ member has root 7388 and is exactly the newly ranked $a(36)$. The structural consequence is derived from the data and the theorem, while the member's rank comes from exhaustive coverage.

Table 6: All observed positive-rate reduced-core classes. Each row satisfies $C(m,\lambda)=S^2$. The count h is restricted to $r\le 10^{10}$; a star marks one of the four inherited families.

m	λ	S	h
9	6	126	12
34	5	396	11
47*	1	148	4
125*	1	500	4
529*	2	2852	7
847	3	7216	7
875*	1	3125	4
1369	2	6586	6
1847	1	5584	4
2209	5	23171	6
6103	6	76862	4
6929	7	102544	3
8750	5	111750	5
28561	1	81289	3
33746	6	445510	3
41503	2	229075	5
230911	6	3076116	2
296009	6	3931172	2
356377	3	2507085	4
371293	5	4227028	2
1067089	2	5059634	3
1159171	5	13891280	2
1830125	1	8984250	3
2724449	3	19985875	3
3457233	4	33665463	2
5182723	1	18966532	2
5185474	2	32708088	3
7140250	3	63163750	2
9020273	5	92443535	1
9759362	2	57933234	3
25574423	2	146048812	2
35701250	4	430062750	1
47316959	1	175360104	2

m	λ	S	h
62344607	1	252382104	2
105971074	3	796214804	1

The table classifies the positive-rate classes represented in this finite dataset. It does not classify all seeds or rates. In particular, the absence of a core from the table does not rule out a family whose smallest member is outside the completed range. The large prime-root count and the rate histogram imply neither an asymptotic density nor an exhaustive list of constructions.

6 Relations and remaining questions

The defining relation uses the square inputs of A000290, the derivative A003415 and its composition A068346 [7, 8, 9]. The paired tables A056798 and A192084 have a different meaning: at their common index j, the first gives an even prime power (including 1) and the second gives its second derivative [13, 14]. Their acquired 10,000 aligned rows contain 72 square-valued second derivatives, with input table ending at 10708731289. Those indices are provider indices; they are not positions in A348425. Moreover, prime-power tables cannot account for mixed roots.

A266890 requires the first arithmetic derivative of a square to be square [15]. Checking $D(n)$ in every consolidated row, using the coverage of Theorem 4.1, gives the finite intersection

$$\mathcal{A} \cap \mathrm{A266890} \cap [0, 10^{20}] = \{0, 1, 4\}.$$

This does not assert a global intersection theorem. For example, $49 \in \mathcal{A}$ but $D(49) = 14$ is not square, whereas 256 has $D(256) = 1024$ square and $D^2(256) = 5120$ not square.

There are also two unconditional exclusions. For $r \geq 2$, $D(r) \geq 1$, and (4) gives $D^2(r^2) \geq r \geq 2$. If such a value is square, it is at least 4, hence neither prime nor squarefree. For $r = 0, 1$, the value is 0, also neither prime nor squarefree. Thus A348425 is disjoint from A192192, whose second derivative is prime, and A328244, whose second derivative is squarefree [16, 17].

The natural structural problem left by Theorem 2.3 is to describe reduced cores m and rates λ for which $C(m, \lambda)$ is square, beyond finite tables of certificates. The prime-power criterion supplies another restricted classification problem as k varies. Neither problem is resolved by the observed counts, and no conjectural density model is used here. The computational question immediately after this work remains the first unresolved index 9115: establishing it requires arbitrary-root coverage beyond 10^{10}, not merely testing another supplied positive witness.

7 Conclusion

The consolidated extension contains 9,114 consecutive terms through input root 10^{10}, with last term 99948010099131627889. The 33-term public snapshot, the two finite continuation stages, and the 8,917-term final campaign have distinct evidential roles. A later complete check closes the original public-prefix dependency; negative coverage of the final interval rests on the single exhaustive producer and its completion records. The next term satisfies (12), while its exact value remains undetermined. The scale criterion turns canonical integer-rate decomposition into provable infinite families, including 35 disjoint classes represented in the data. It explains a structured part of the mixed-root behavior without converting a finite census into a global classification.

Data availability

The public project repository and Zenodo archive are identified in [24]. The publication package contains the indexed terms, documentary witness table, normalized C source and build instructions, original execution materials, provenance checks, Python validators, monolithic manuscript source, finite-analysis companion, and matching PDF. The cited Zenodo DOI identifies the project series; individual archived releases have distinct version-specific records. Original source reviews remain internal materials. Complete independent re-enumeration of the final search interval is not supplied.

AI use disclosure

Large language models were used as research and drafting assistants during the A348425 workflow, including OpenAI Codex and external LLM-based review systems from the ChatGPT, Claude, DeepSeek, Gemini, Grok, Qwen, MiniMax, Meta AI, GLM and Mistral families. These are the product or family names present in the project records; exact model-version metadata were not consistently available or authenticated. The retained history spans October 5–23, 2026; the first manuscript review was integrated on October 23. These channels are not claimed to be independent providers.

Mathematical analysis. OpenAI Codex assisted with literature research, product-rule derivations, canonical-core analysis, and the distinction between indexed terms and unindexed witnesses. The four inherited families were already present in project notes before the final campaign. The manuscript's additional class certificates and tightened bound were checked by exact arithmetic on retained data and primary sources, not by accepting conversational assertions as evidence.

Algorithm and code. OpenAI Codex assisted with drafting the C implementation, validation planning, and integration of supplied code critiques. Claude-supplied Python and C implementations were locally rerun for the finite continuation and bounded revision checks, respectively; they do not constitute independent full-campaign reproduction. Such assistance and code review are fallible and do not replace the proofs, executable checks, or the stated coverage boundaries. The author ran the official campaign. No personal verification of every proof or manual recalculation of every output row is asserted.

Manuscript. OpenAI Codex assisted with drafting, source reconciliation, and internal checking of this text. The executable validators, result tables, original run records and primary sources form the verification basis. Carlo Corti retains full scientific responsibility for the mathematical statements, implementation, numerical results, validation boundaries, and conclusions.

A Build and completion-record specification

The executed Makefile uses `gcc`, with `-O3 -std=c17`, `-Wall`, `-Wextra`, `-Wpedantic`, `-Wshadow`, `-Wconversion`, `-Wdouble-promotion`, `-Wformat=2`, `-Werror`, and `-fopenmp`; linking uses `-fopenmp -lcrypto -lm`. Two preprocessor string definitions bind the source and Makefile identities:

`A348425_SOURCE_SHA`, `A348425_MAKE_SHA`.

The original build file, rather than a reconstruction of its shell quoting, is retained with the execution materials.

The configuration starts with `A348425-CFG-1` and records `start`, `end`, `block`, `group`, `threads`, and the source, Makefile and executable bindings. The executed values are 1000001, 10^{10}, 65536, 2 and 16; `group` is the number S of segments. More precisely, let u, v be the configured inclusive root endpoints and ℓ the block size. There are $K = \lfloor (v-u)/\ell \rfloor + 1$ blocks. Define

$$q_j = j + \left\lfloor \frac{(K-S)j(6S+1-j)}{S(5S+1)} \right\rfloor, \qquad 0 \le j \le S.$$

Segment j, numbered from 0, covers $[u + q_j\ell,\ \min(v, u + q_{j+1}\ell - 1)]$ for $0 \le j < S$. This reproduces the boundaries in Table 2; $q_0 = 0$ and $q_S = K$, and the reserved block in each positive increment $q_{j+1} - q_j$ ensures nonempty contiguous segments.

A segment starts with `A348425-SEG-1`, followed by configuration binding, segment identifier, inclusive `low` and `high`, `next`, internal `elapsed_ns`, and `hits`. Its positive records have five tab-separated unsigned decimal columns $(r, r^2, D(r^2), D^2(r^2), R)$ in strictly increasing r. The trailing `state_sha` binds every preceding byte of that segment; a separate seal binds the full segment, including this trailing line. Segment filenames are `sNNNN.tsv`, with four lowercase hexadecimal digits for the segment identifier; the matching external seal file contains 64 lowercase hexadecimal digits followed by LF. The manifest `index.tsv` begins with `A348425-MANIFEST-1`, a TAB, and `cfg_sha=` followed by the configuration digest. Its second line has the TAB-separated labels `id`, `low`, `high`, `state_sha`, `elapsed_ns`, `hits`, in that order. The manifest column labelled `state_sha` stores the digest of the *entire* segment file, equal to its external seal; it is distinct from the in-segment field of the same name, which covers only the preceding segment bytes. Thus the ordered manifest binds the configuration and the full segment files, including their trailing digest lines.

The aggregate `hits.tsv` begins with `A348425-HITS-1`, a TAB, and `cfg_sha=` followed by the configuration digest. Its second line has the TAB-separated column labels `r`, `n`, `x`, `y`, `R`, corresponding to the five coordinates above. Every record line ends in LF. A completed segment must have `next` = `high` + 1; the next segment starts at that cursor. The aggregate hit list is regenerated from the segments and checked for equality, rather than admitted as a separate authority. These conditions expose missing, reordered or inconsistent records. They establish integrity and recorded completion; the exhaustive algorithm remains the basis for negative coverage.

The two saved internal elapsed counters correspond to approximately 1539 and 1387 seconds, respectively; their full precision is retained in the execution records. Their sum has the timing limitations stated in Section 3.2. The finite-analysis companion requires SymPy 1.12, deliberately rejecting other versions to preserve its tested environment. It can be run from an isolated Python environment with that version installed; it checks positive rows and class certificates, not exhaustive negative coverage.

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